

VASILEIOS KARAGIANNIDIS

Junior Software / Embedded Engineer · Robotics Researcher

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PROFESSIONAL SUMMARY

Final-year Computer Engineering student with hands-on experience in Python, PyTorch, embedded systems (ESP32, Raspberry Pi) and active research work in robotics and deep learning. My bachelor thesis involves the development of a DIY robotic arm built around a Raspberry Pi 5 + 2× Pico and multiple stepper motors, while I'm also working on computer vision and mobile manipulation with the Boston Dynamics Spot robot on NVIDIA Isaac Sim. I have additionally completed an academic research project comparing multimodal LLMs with fine-tuned Vision Transformers for medical imaging. Vice President of the IEEE IHU Student Branch and co-organizer of the IEEEetcon 2026 conference.

Looking for: Internship or Junior position in Software Development, Embedded Systems, Robotics or IoT teams — full-time, on-site/hybrid in Thessaloniki or remote.

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript, SQL, C/C++ (embedded)

Frameworks & Libraries: JavaFX, PyTorch, HuggingFace Transformers, CustomTkinter, Tkinter, scikit-learn, pandas, PIL

Databases & Cloud: PostgreSQL, JDBC, Firebase (real-time DB & cloud sync)

Machine Learning: Vision Transformers (ViT), Multimodal LLMs (LLaVA), fine-tuning, evaluation metrics, ISIC 2017

Embedded & Hardware: Raspberry Pi 5, Raspberry Pi Pico, ESP32, stepper motors, circuit design & analysis

Robotics & Simulation: NVIDIA Isaac Sim, Boston Dynamics Spot SDK, Forward/Inverse Kinematics, Computer Vision, Mobile Manipulation

Tools & Platforms: Git, GitHub, Linux, VS Code, PhotoRec, file system automation

Areas of Interest: Robotics, Deep Learning, IoT, Embedded Systems, Computer Vision, Automation

BACHELOR THESIS

DIY Robotic Arm — International Hellenic University 2025 – 2026 (in progress)

Design and development of a DIY robotic arm capable of stably lifting objects from 500g up to 2kg, with the sensitivity required to grip an egg without breaking it.

- Working on the arm's kinematics — searching for the optimal approach (Forward / Inverse Kinematics)
- Hardware: multiple stepper motors controlled by a Raspberry Pi 5 and 2× Raspberry Pi Pico
- Developing control software and coordinating communication between the controllers
- Goal: stability + sensitivity across a wide range of object weights and fragility

PROFESSIONAL & ACADEMIC EXPERIENCE

Researcher — Robotic Simulation & Vision — International Hellenic University 2025 – Present

- Set up and developed a simulation environment in NVIDIA Isaac Sim for the Boston Dynamics Spot quadruped robot
- Working on the Spot vision system — extracting and processing visual data from the robot's sensors
- Implementing mobile manipulation tasks: autonomous navigation to a target and object handling via the robotic arm (pick & place)
- Continuous deepening in the simulation environment, configuration and robotic application development workflow

Vice President — IEEE IHU Student Branch 2025 – 2026

- Co-organizer of the two-day IEEEetcon 2026 conference — inviting and coordinating speakers, liaising with companies and academic teams, securing sponsorships

- Co-organized the branch's participation in the international IEEE Extreme competition (October 2025) — secured sponsorships and achieved a high final ranking
- Designed, organized and delivered technical workshops for students: 'Introduction to Microcontrollers' and 'Microcontrollers & Robotics'

Kitchen Coach — KFC Thessaloniki

2020 – 2025

- Training and mentoring of new kitchen staff — onboarding & coaching
- Kitchen team supervisor and shift workflow management
- Mediator for store-level operational issue resolution
- Enforcement of Food Safety standards

SELECTED PROJECTS

Multimodal Medical Image Classification — Academic research project: comparison of zero-shot multimodal LLM (LLaVA) against a fine-tuned Vision Transformer (ViT) on dermatological image classification (ISIC 2017). The fine-tuned ViT achieved 78% accuracy — significantly higher than the zero-shot LLaVA (39%). Documented in an IEEE-format paper.

Tech: Python, PyTorch, HuggingFace Transformers, ViT, LLaVA, ISIC 2017

Security-System-with-ESP32 — ESP32-based security system — sensors, real-time alerts and control.

Tech: ESP32, C/C++, sensors, embedded

Email Automation Suite — Two Python email automation tools — one for speaker/sponsor communication at IEEEetcon (automailforieee) and one for cold outreach to tech companies (automailtech).

Tech: Python, smtplib, CSV processing, email automation

youtube-to-mp3 — Desktop app for quick YouTube-to-MP3 conversion with a clean, user-friendly interface.

Tech: Python, yt-dlp, Tkinter (GUI)

movie-planner — Cloud-based watchlist app with real-time sync — personal movie list featuring a modern GUI.

Tech: Python, Firebase (cloud DB), CustomTkinter

Fast-Food-App — Order management application for a fast-food store with persistent storage — inspired by my KFC experience.

Tech: Java, JavaFX, PostgreSQL, JDBC

Data Recovery & Photo Organizer — Python tool for file recovery and automatic organization after data loss — grouping photos by EXIF metadata, duplicate detection, and date-based folder structure.

Tech: Python, PIL/Pillow, EXIF metadata, PhotoRec, file system automation

All projects: github.com/karagiancode

EDUCATION

International Hellenic University (IHU) — Thessaloniki

2019 – Present

Department of Information & Electronic Systems Engineering — Specialization: Computer Science & Programming

30th General Lyceum of Thessaloniki

2016 – 2019

High School Diploma

CERTIFICATIONS, LANGUAGES & MORE

- Certifications: International Diploma in IT Skills (Cambridge) · Basic Life Support / BLS (ERC)
- Languages: Greek (native) · English (B2 — good command)
- Volunteering: Upcoming volunteer work with the TechSaloniki organization (2026)
- Soft skills: Leadership, teamwork, working under pressure, communication, eager to learn and adapt
- Interests: Tae Kwon Do (WTF, 2013 – 2018), Chess, Gymnastics